

SOFTWARE ENGINEERING LAB

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

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Topic	Telemedicine Slot Booking & Prescription Portal
Module	Set 2: Payment & Booking Changes (UC-03 Make Payment, UC-08 Cancel / Reschedule Slot)
Jira space	Telemedicine Payment - Harsha (key: TPH), Scrum board

1. Introduction

This report documents the Jira work for Lab 2. The functional requirements of my Lab 1 module (Set 2: Payment & Booking Changes) were converted into Epics and User Stories, prioritised, estimated with Fibonacci story points, and delivered through three one-week sprints on a Scrum board. The three Epics are specific to my module and cover the payment, cancellation and rescheduling use cases of the Telemedicine Slot Booking & Prescription Portal.

Source requirements from Lab 1: UC-03 Make Payment, UC-08 Cancel / Reschedule Slot (UC-08 «extend» UC-02, UC-02 «include» UC-03), FR-003 (exclusive slot reservation once payment is confirmed) and NFR-001 (encryption of data in transit and at rest). The Payment Gateway is an external actor, and the cut-off window and refund behaviour come from the UC-08 flow specification.

2. Epics

Epic	Description	Lab 1 source	Stories	Points
Epic 1: Secure Consultation Payment	Enable patients to pay for a consultation through the external payment gateway and have their slot reserved exclusively once payment is confirmed.	UC-03, FR-003, NFR-001	5	23
Epic 2: Appointment Cancellation & Refunds	Allow patients to cancel confirmed bookings within the clinic's policy window and receive refunds through the payment gateway.	UC-08 (cancel path)	5	24
Epic 3: Appointment Rescheduling	Let patients move a confirmed booking to a new slot with the same physician without paying again.	UC-08 (reschedule path)	4	18
Total			14	65

3. User Stories, Priority and Story Points

Each story follows the format “As a [role], I want [goal], so that [benefit]”. Priority reflects user impact and risk: payment confirmation, exclusive slot reservation and refunds are the most critical because they involve money and prevent double-booking. Story points use the Fibonacci scale (1, 2, 3, 5, 8, 13).

Key	Story	Description	Priority	Points
TPH-4	Story 1.1: Payment Summary	As a patient, I want to see the consultation fee, taxes and slot details before paying, so that I know exactly what I am being charged.	High	3
TPH-5	Story 1.2: Pay via Gateway	As a patient, I want to pay securely via UPI, card or net banking through the payment gateway, so that I can confirm my booking online.	Highest	8
TPH-6	Story 1.3: Exclusive Slot Reservation	As a patient, I want my slot reserved exclusively for me once payment is confirmed, so that no other patient can double-book it.	Highest	5
TPH-7	Story 1.4: Failed Payment Handling	As a patient, I want to be told clearly when a payment fails or times out and be able to retry, so that I do not lose my slot or get charged twice.	High	5
TPH-8	Story 1.5: Receipt & Booking Reference	As a patient, I want a payment receipt and booking reference, so that I have proof of payment.	Medium	2
TPH-9	Story 2.1: My Appointments & Policy	As a patient, I want to view my upcoming bookings with the cancellation and refund policy, so that I can make an informed decision.	High	3
TPH-10	Story 2.2: Cancel Booking	As a patient, I want to cancel a confirmed upcoming booking, so that I am not committed when my plans change.	High	5
TPH-11	Story 2.3: Automatic Refund	As a patient, I want eligible refunds initiated automatically to my original payment method per the policy tier, so that I get my money back without chasing support.	Highest	8
TPH-12	Story 2.4: Cut-off Window Enforcement	As a patient, I want to be told when my request falls inside the no-change window (under 2 hours) and see the non-refundable amount, so that I can keep the booking or cancel without refund.	High	5
TPH-13	Story 2.5: Physician Notification	As an attending physician, I want to be notified when a patient cancels, so that I can use the freed time.	Medium	3
TPH-14	Story 3.1: Choose New Slot	As a patient, I want to pick a new available slot with the same physician, so that I can keep my consultation at a convenient time.	Highest	8
TPH-15	Story 3.2: Reuse Existing Payment	As a patient, I want the new slot reserved against my existing payment, so that I do not have to pay twice.	High	5
TPH-16	Story 3.3: Release Original Slot	As a patient searching for slots, I want a rescheduled slot to become available again, so that I can book it.	Medium	3
TPH-17	Story 3.4: Reschedule Confirmation	As a patient, I want a confirmation with the updated booking reference, and the physician notified, so that both of us know the new time.	Medium	2
Total				65

Estimation approach. Because I worked alone, planning poker was replaced by relative sizing: the smallest story (receipt, 2 points) was the baseline, and every other story was compared against it for complexity, amount of work and uncertainty. The gateway payment

and automatic refund stories received 8 points because they depend on an external system, involve money, and have several failure cases.

4. Jira Backlog with Epics and User Stories

Figure 1 shows the complete product backlog with all 14 stories grouped under the three Epics before any sprint was started. Figures 2 to 4 filter the backlog by Epic. The grey box on each story row is its story point value, and the arrow icon is its priority. The header of each filtered backlog shows the Epic total (23, 24 and 18 points).

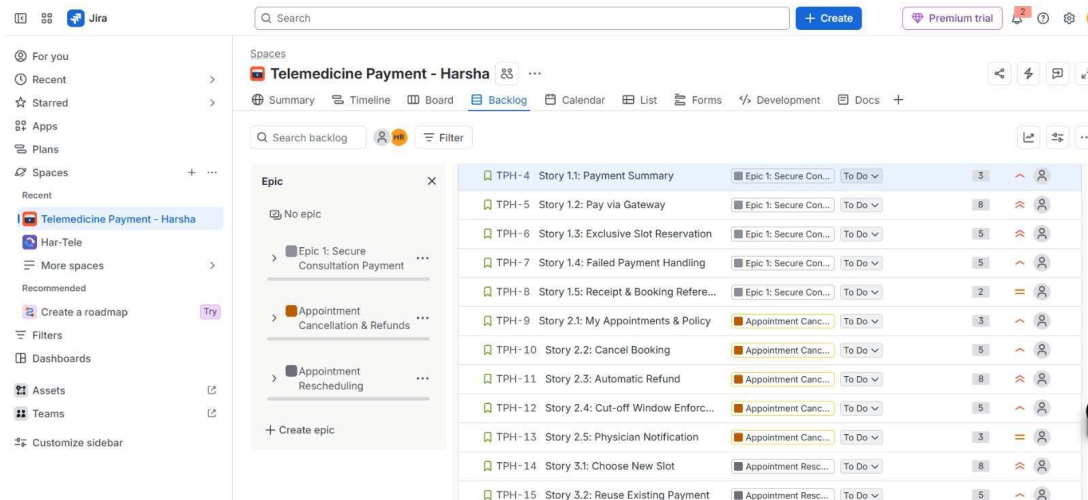


Figure 1: Product backlog with Epics (left panel) and all 14 user stories, priority and story points

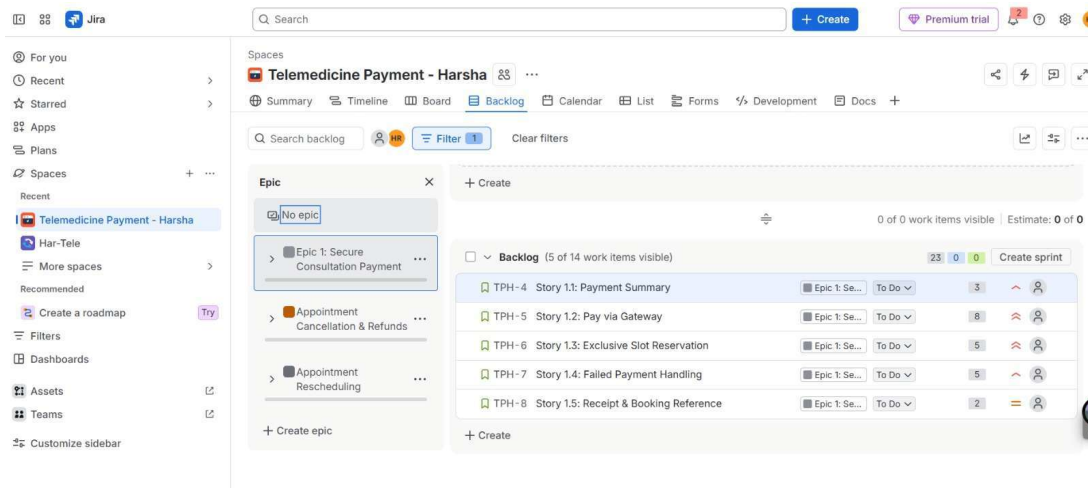


Figure 2: Epic 1 (Secure Consultation Payment): 5 stories, 23 story points

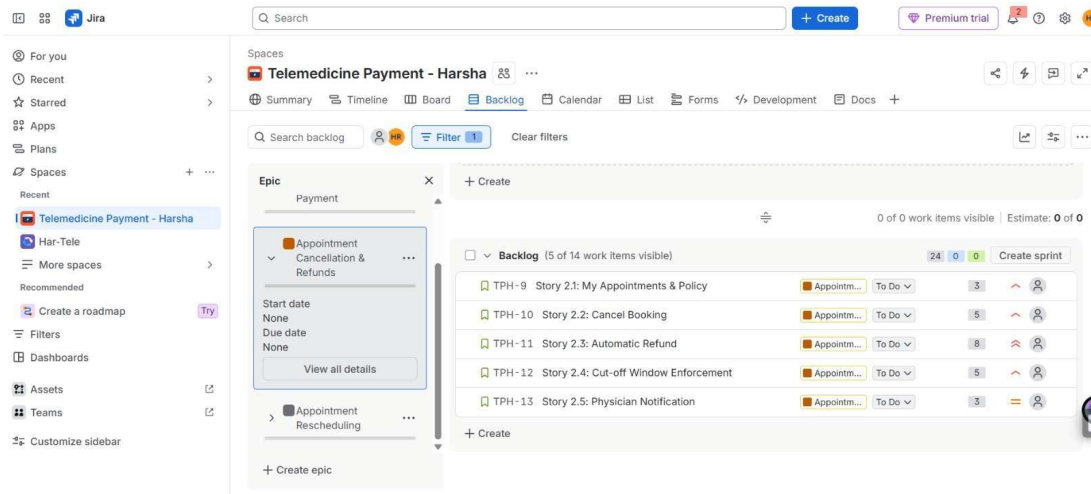


Figure 3: Epic 2 (Appointment Cancellation & Refunds): 5 stories, 24 story points

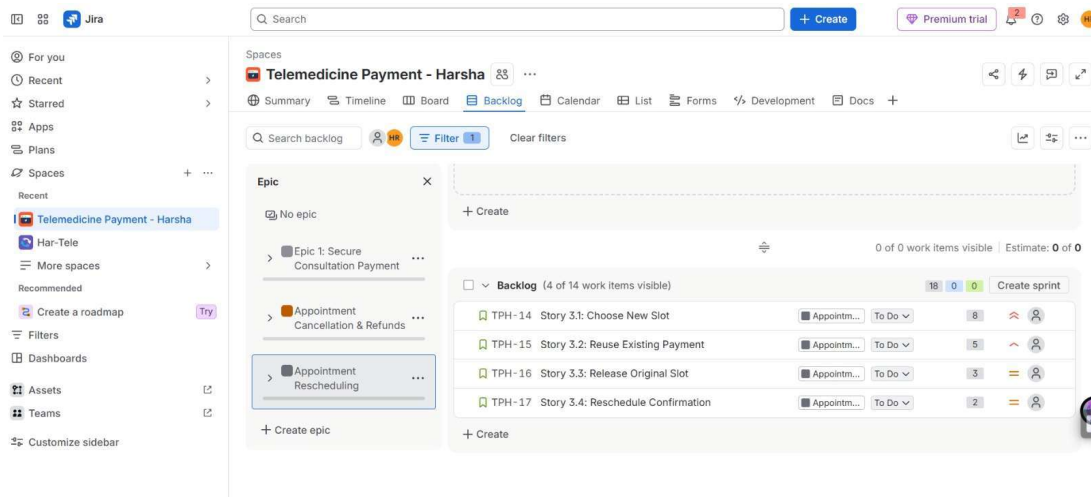


Figure 4: Epic 3 (Appointment Rescheduling): 4 stories, 18 story points

5. Story Point Assignments

The story point values are visible in the Figures 1 to 4 screenshots above and are summarised by Epic below.

Epic	Stories	Story points	Priority mix
Epic 1: Secure Consultation Payment	5	$3 + 8 + 5 + 5 + 2 = 23$	2 Highest, 2 High, 1 Medium
Epic 2: Appointment Cancellation & Refunds	5	$3 + 5 + 8 + 5 + 3 = 24$	1 Highest, 3 High, 1 Medium
Epic 3: Appointment Rescheduling	4	$8 + 5 + 3 + 2 = 18$	1 Highest, 1 High, 2 Medium
Total	14	65	

6. Sprint Simulation

Work was planned as three one-week sprints, each delivering one Epic. In every sprint the stories were moved from To Do to In Progress to Done in priority order.

Sprint	Dates	Sprint goal	Stories	Committed	Completed
TPH Sprint 1	2 Oct to 9 Oct 2026	Patient can pay and get a confirmed slot	5	23	23
TPH Sprint 2	3 Oct to 10 Oct 2026	Patient can cancel and receive a refund per policy	5	24	24
TPH Sprint 3	3 Oct to 10 Oct 2026	Patient can reschedule a confirmed booking to a new slot with the same physician without paying again	4	18	18

6.1 Sprint 1: Epic 1 (Secure Consultation Payment)

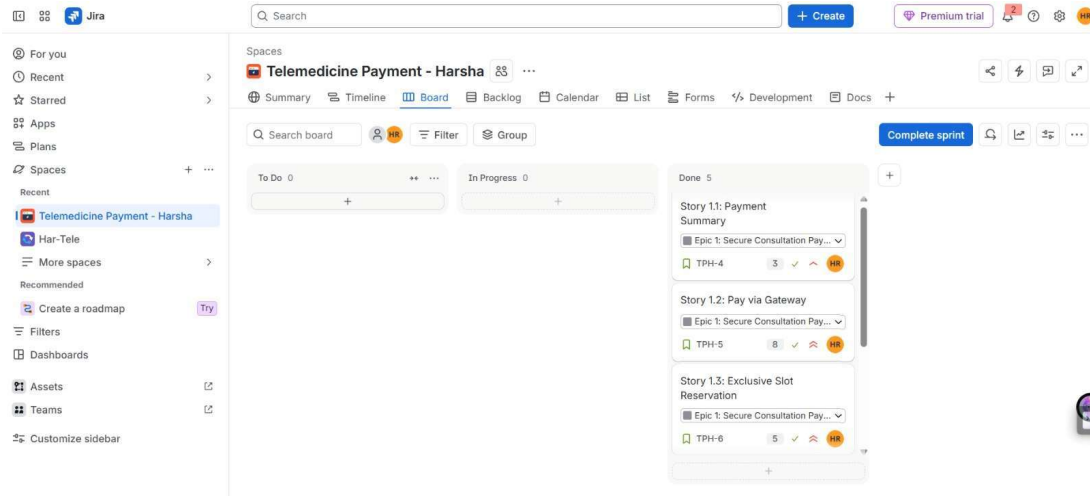


Figure 5: Sprint 1 board (Active sprint view): all 5 stories in Done

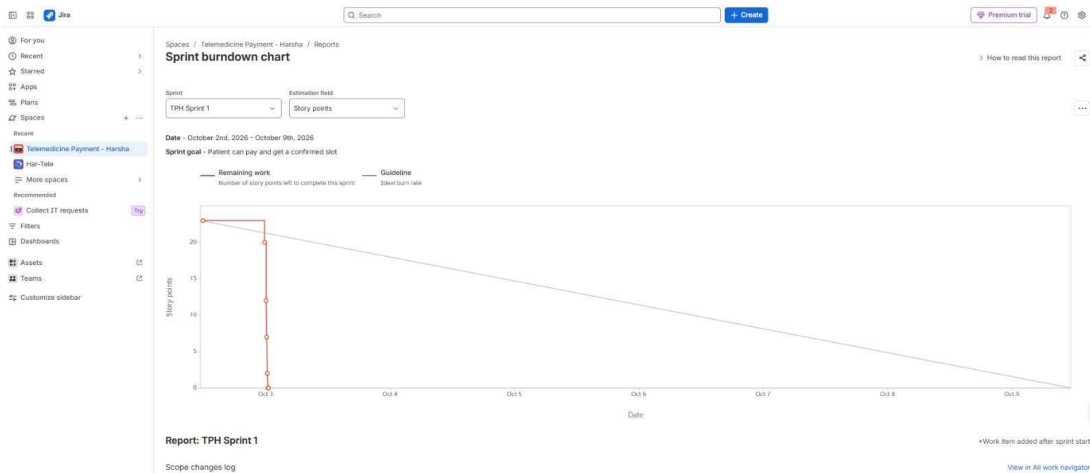


Figure 6: Sprint 1 burndown chart (story points)

Remaining points after each completed story: $23 \rightarrow 20 \rightarrow 12 \rightarrow 7 \rightarrow 2 \rightarrow 0$.

6.2 Sprint 2: Epic 2 (Appointment Cancellation & Refunds)

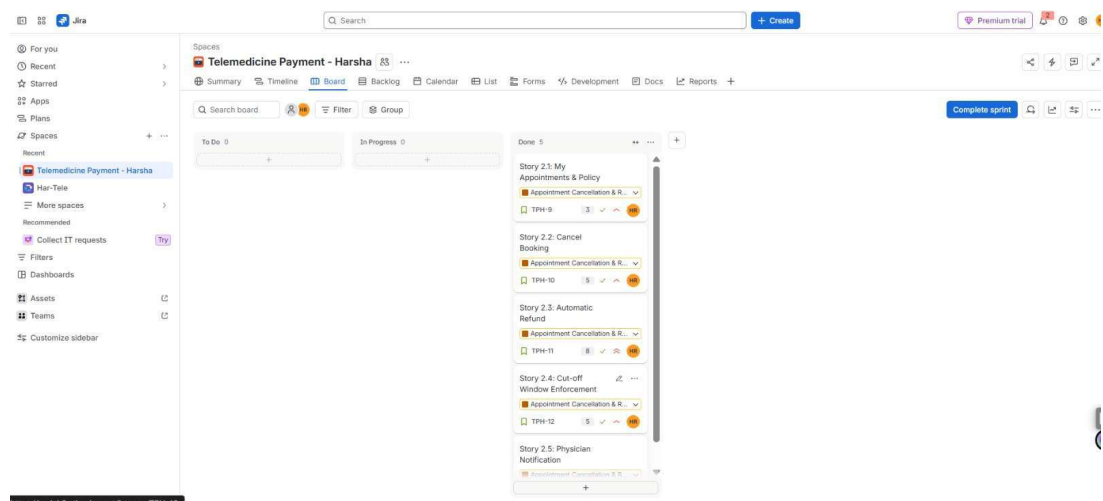


Figure 7: Sprint 2 board (Active sprint view): all 5 stories in Done

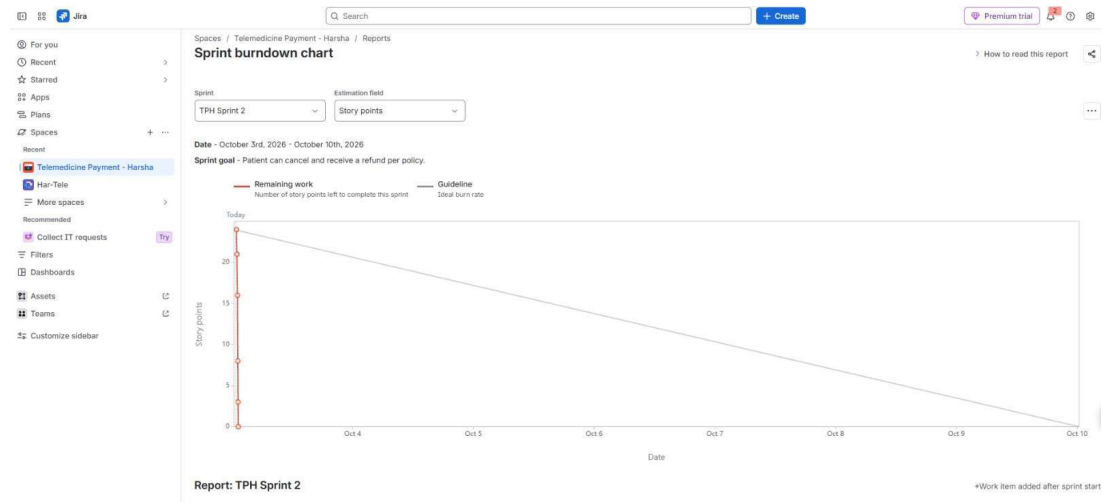


Figure 8: Sprint 2 burndown chart (story points)

Remaining points after each completed story: 24 → 21 → 16 → 8 → 3 → 0.

6.3 Sprint 3: Epic 3 (Appointment Rescheduling)

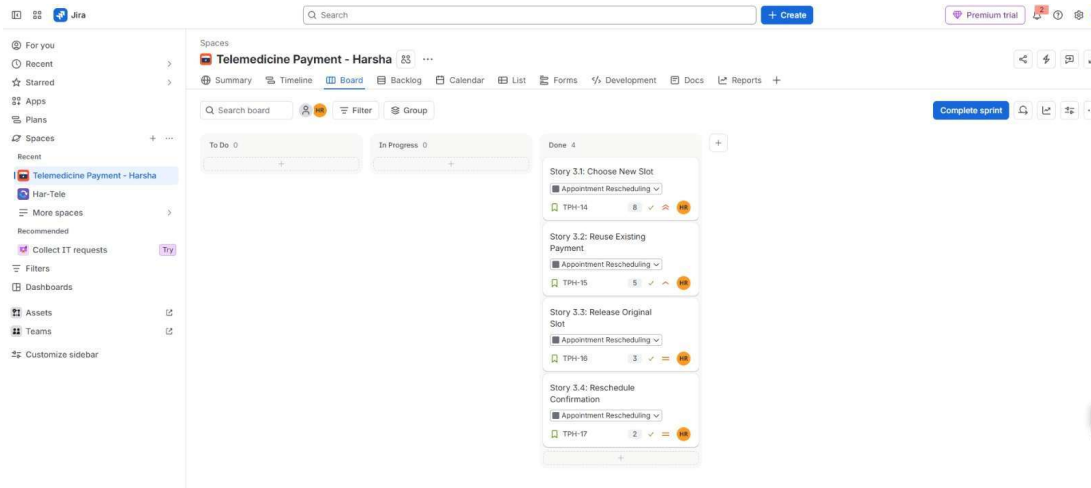


Figure 9: Sprint 3 board (Active sprint view): all 4 stories in Done

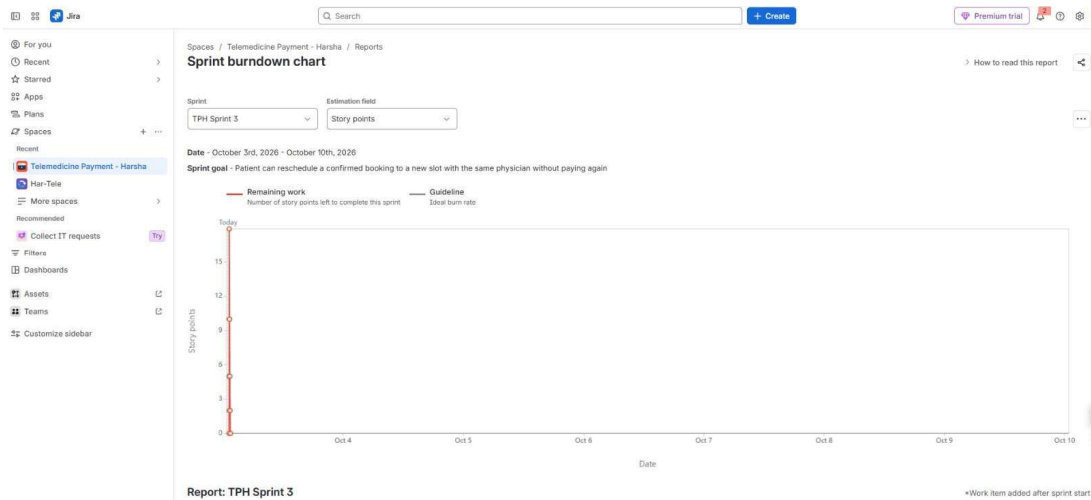


Figure 10: Sprint 3 burndown chart (story points)

Remaining points after each completed story: $18 \rightarrow 10 \rightarrow 5 \rightarrow 2 \rightarrow 0$.

6.4 Backlog after all sprints

After Sprint 3 was completed, all three Epic progress bars are fully green and the backlog is empty (0 work items). Jira automatically created an empty TPH Sprint 4 ready for future work.

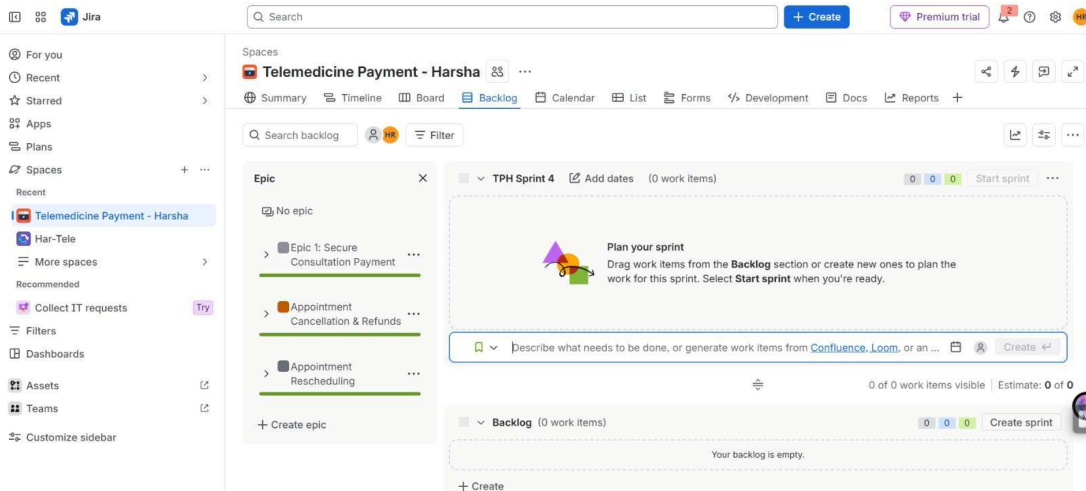


Figure 11: Backlog after Sprint 3: all Epics complete, backlog empty

6.5 Sprint summary

Sprint	Committed points	Completed points	Carry-over	Velocity
TPH Sprint 1	23	23	0	23
TPH Sprint 2	24	24	0	24
TPH Sprint 3	18	18	0	18
Total	65	65	0	Average 21.7

7. Reflection

1. Did your estimations reflect the actual effort?

Partly. The relative sizes were sensible: the stories that depend on the external payment gateway or involve refunds (Stories 1.2, 2.3 and 3.1, 8 points each) were clearly the heaviest, and small stories such as the receipt (2 points) were the lightest. All three sprints finished with every committed point delivered, so the estimates were never too optimistic. However, because this was a simulation, the cards were moved to Done by me rather than built by a team, so “actual effort” was not measured in real working time. In a real project, Story 1.4 (Failed Payment Handling, 5 points) would probably deserve 8 points, because timeouts and duplicate-charge prevention add hidden complexity. The Fibonacci scale helped separate small, routine work from risky work.

2. Was your backlog well-prioritised?

Yes. The stories on the critical path (pay via gateway, exclusive slot reservation, automatic refund and choosing a new slot) were set to Highest because they protect revenue, prevent double-booking (FR-003) and carry financial risk. Convenience stories such as the receipt and the confirmation messages were Medium. The Epic order also respects the dependencies in my Lab 1 use-case diagram: a booking must be paid for (UC-02 «include» UC-03) before it can be cancelled or rescheduled (UC-08 «extend» UC-02), so payment was delivered in Sprint 1, cancellation and refunds in Sprint 2 and rescheduling in Sprint 3. No story had to be reordered during the sprints.

3. How did your simulated sprint align with your plan?

The sprints followed the plan closely. In each sprint all committed stories were moved from To Do to In Progress to Done in priority order, no stories were added or removed after the sprint started, and nothing was carried over (committed and completed points were 23/23, 24/24 and 18/18). The main difference from a real sprint is timing: each one-week sprint was finished within roughly a day, because the work was simulated and I could move cards immediately. A real team would spread this work across the week and would encounter blockers, rework and changes of scope that this simulation does not show.

4. What insights did the burndown chart give about your team's capacity?

In all three charts the red remaining-work line falls in steps (for Sprint 1: 23, 20, 12, 7, 2, 0) and ends far below the grey guideline, which shows that the work was completed well ahead of the ideal burn rate. The very steep drop would suggest, in a real team, that the sprints were under-committed or that stories were closed in a batch instead of steadily through the week. The velocity was 23, 24 and 18 points, an average of about 22 points per sprint, which is a useful baseline for planning. Since the team finished early every time, the next sprint could take on more work, for example 25 to 30 points. The charts also show that scope stayed stable, because the starting total never changed after the sprint began.